

Investo

Smarter Investing Made Simple | Team Contributions & Division of Labor

In accordance with the hackathon submission guidelines, this document certifies a balanced and equal division of labor and technical achievements for the Investo project. The development of Investo was divided into 4 key product pillars: Backend & Architecture, Data Engineering & Scraping, Frontend Views & Native Mobile, and Product Design & Simulation Interfaces. Each of the 4 team members spearheaded one of these pillars to deliver a complete, high-fidelity end-to-end agentic solution.

1. Zainab Fatima (Lead Full-Stack & System Architect)

Role: Backend Architecture, System Flow, and Integration

- Engineered the Express.js server foundation, setting up robust REST APIs and SSE endpoints.
- Designed and programmed the central authentication and JWT-based session vault (authService.js).
- Built the structured ingestion/normalization schemas mapping diverse platform data layers.
- Developed backend simulation run routines for contradictory, budget, and rollback test profiles.
- Implemented security controls protecting external API routes and preventing local memory leaks.

2. Muhammad Ali (Data & Scraper Engineer)

Role: Crawlers, Ingestion, and Contradiction Resolution

- Coded target crawlers pulling real-time property listings from local portals (Zameen, Graana).
- Designed the Anomaly Credibility Weighting algorithm resolving pricing conflicts.
- Built data cleaning pipelines filtering out duplicate listing noise and incomplete feeds.
- Integrated economic fallbacks (e.g. mutual funds, bonds) when property yields fail targets.
- Setup API request rate limiting mechanisms preventing crawler blocks by remote platforms.

3. Ayesha Khan (Frontend Engineer & Capacitor Specialist)

Role: Onboarding Forms, Results Viewer, and Native Mobile Builds

- Built the interactive investment onboarding profiles with range selectors and risk tolerances.
- Developed the dynamic tabs view for results, data sources, contradiction ledger, and trace logs.
- Setup local storage synchronizers ensuring session state persistence upon window reloads.
- Orchestrated the Capacitor native build settings creating functional Android/iOS project shells.
- Handled form inputs validation, error catches, and loading state visuals during agent scoring.

4. Bilal Ahmed (UI/UX Designer & Product Manager)

Role: Branding, Marketing Hero, and Interactive Simulations

- Overhauled brand identity, naming, and taglines focusing on ease of investment for users.
- Designed the glowing glassmorphic headers and high-contrast typography gradients on the hero section.
- Programmed the dynamic CSS Live Terminal Simulator illustrating Karachi agent crawling loops.
- Overhauled the Simulation Sandbox, converting demo triggers into high-fidelity hover-cards.
- Coordinated markdown documentation (README, walkthroughs, tasks) and trace packages for submission.

Certification of Equal Work Contribution

We, the undersigned, certify that the work for the Investo project was divided equally, collaboratively, and fairly. Every member

fully contributed to their core responsibilities to create the complete, state-of-the-art agentic application submitted for evaluation.

Zainab Fatima (Lead Architect) _____

Muhammad Ali (Scraper Engineer) _____

Ayesha Khan (Frontend & Capacitor) _____

Bilal Ahmed (UI/UX & PM) _____